

RayEncoder Evaluation Report

Checkpoint: checkpoint/003_RayEncoder/checkpoint/ray_encoder_best.pth

Split: val | Samples: 128

8-way direction accuracy: 42.97%

Ray cosine: mean 0.8216, median 0.8628, p10 0.5000, p90 1.0000

Angular error: mean 38.92 deg, median 42.82 deg, p90 90.00 deg

Confidence: mean 0.9089, median 0.9882

Confusion Matrix

Rows are ground truth, columns are predicted. Diagonal cells are correct.

		Predicted direction							
		N	NE	E	SE	S	SW	W	NW
Ground Truth	N	9 38%	1 4%	0 0%	2 8%	0 0%	4 17%	3 12%	5 21%
	NE	2 17%	4 33%	3 25%	3 25%	0 0%	0 0%	0 0%	0 0%
	E	0 0%	1 6%	5 29%	6 35%	3 18%	0 0%	1 6%	1 6%
	SE	0 0%	2 17%	1 8%	8 67%	1 8%	0 0%	0 0%	0 0%
	S	0 0%	1 6%	1 6%	1 6%	9 53%	5 29%	0 0%	0 0%
	SW	0 0%	0 0%	0 0%	1 7%	0 0%	9 64%	4 29%	0 0%
	W	0 0%	0 0%	0 0%	0 0%	0 0%	3 23%	5 38%	5 38%
	NW	4 21%	0 0%	0 0%	1 5%	0 0%	3 16%	5 26%	6 32%

Best 5 Samples by Angular Error

Input image



scene: Image054
GT: S (180 deg)
Pred: S (180.0 deg)
angle error: 0.0 deg
ray cosine: 1.0000
confidence: 1.000

Input image_5500.png



scene: Image142
GT: NW (315 deg)
Pred: NW (315.0 deg)
angle error: 0.0 deg
ray cosine: 1.0000
confidence: 1.000

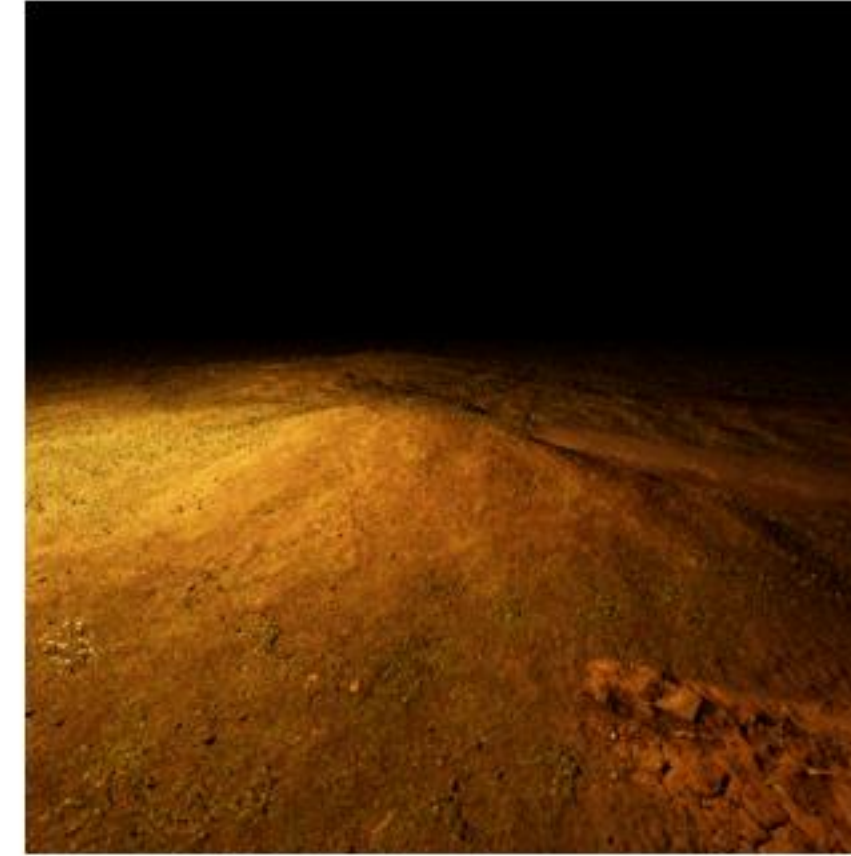
Input image_NW_3500.png



scene: Image142
GT: E (90 deg)
Pred: E (90.0 deg)
angle error: 0.0 deg
ray cosine: 1.0000
confidence: 1.000

Image142_E_2500.png

Input image



scene: Image292
GT: W (270 deg)
Pred: W (270.0 deg)
angle error: 0.0 deg
ray cosine: 1.0000
confidence: 1.000

Input image_W_2500.png



scene: Image003
GT: SE (135 deg)
Pred: SE (135.0 deg)
angle error: 0.0 deg
ray cosine: 1.0000
confidence: 1.000

Image003_SE_2500.png

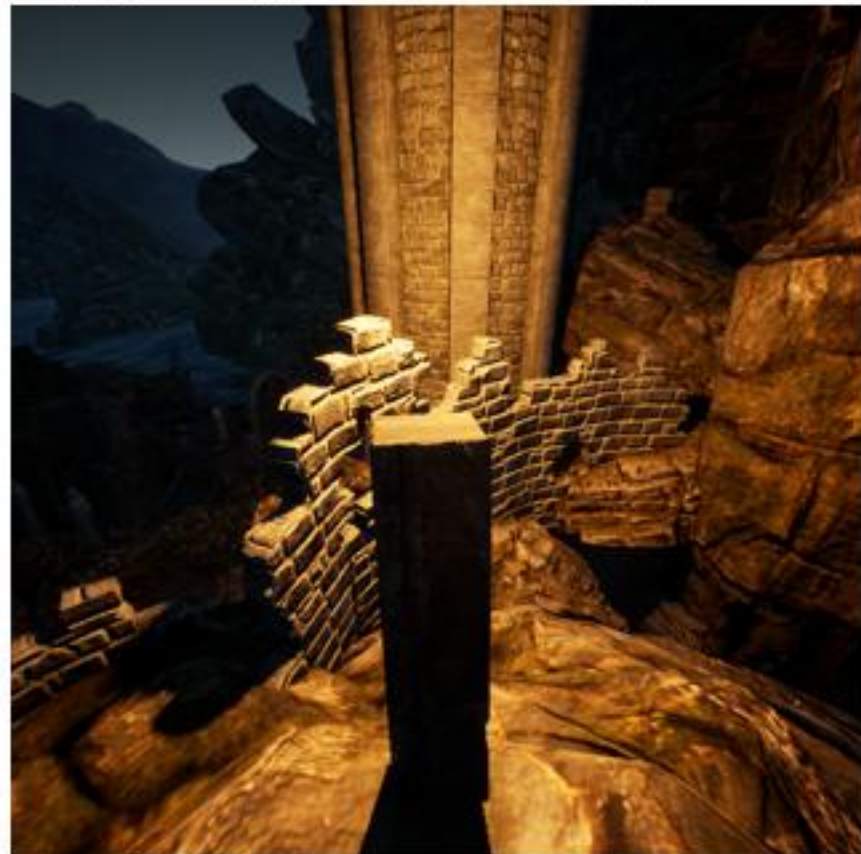
Worst 5 Samples by Angular Error

Input image



scene: Image151
GT: NW (315 deg)
Pred: SE (134.9 deg)
angle error: 179.9 deg
ray cosine: 0.0000
confidence: 0.998

Input image_W_4500.png



scene: Image127
GT: N (0 deg)
Pred: SW (224.3 deg)
angle error: 135.7 deg
ray cosine: 0.1423
confidence: 0.988

Input image_2500.png



scene: Image088
GT: N (0 deg)
Pred: SE (134.8 deg)
angle error: 134.8 deg
ray cosine: 0.1475
confidence: 0.994

Image088_N_2500.png

Input image



scene: Image034
GT: E (90 deg)
Pred: W (270.3 deg)
angle error: 179.7 deg
ray cosine: 0.0000
confidence: 0.992

Input image_6500.png



scene: Image057
GT: N (0 deg)
Pred: SW (225.0 deg)
angle error: 135.0 deg
ray cosine: 0.1462
confidence: 0.999

Image057_N_2500.png

Heatmap Analysis 1: scene Image240

Each panel is the same scene under a different GT light direction. Heatmap target: predicted direction logit.



GT: N (0 deg)
Pred: N (1.4 deg)
error: 1.4 deg
cos: 0.9999 conf: 0.964

Image240_N_4500.png



GT: NE (45 deg)
Pred: NE (44.9 deg)
error: 0.1 deg
cos: 1.0000 conf: 0.999

Image240_NE_4500.png



GT: E (90 deg)
Pred: NE (36.8 deg)
error: 53.2 deg
cos: 0.7994 conf: 0.781

Image240_E_6500.png

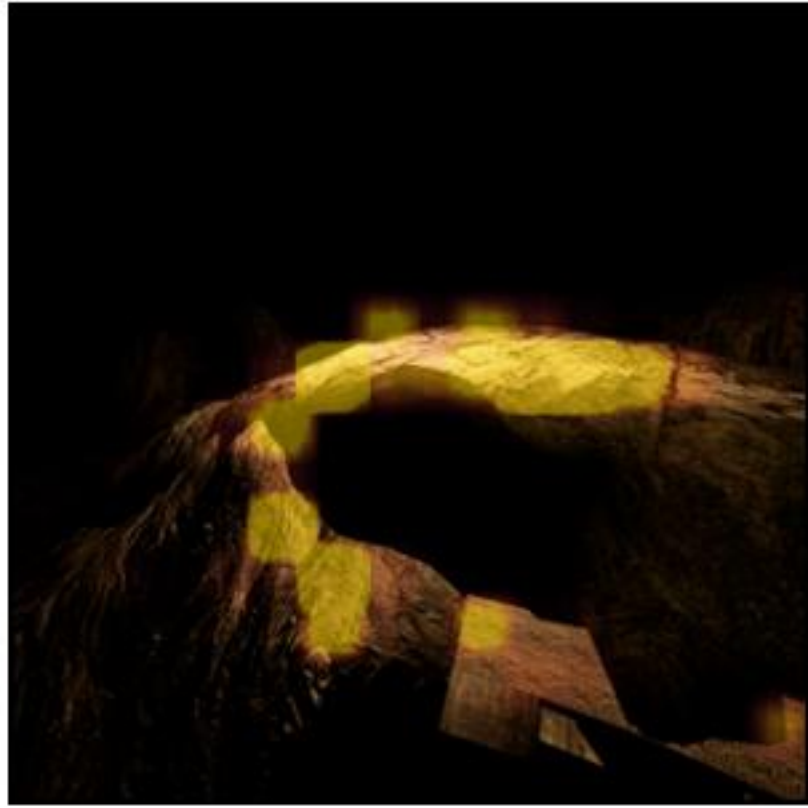


GT: S (180 deg)
Pred: SW (225.0 deg)
error: 45.0 deg
cos: 0.8536 conf: 0.999

Image240_S_5500.png

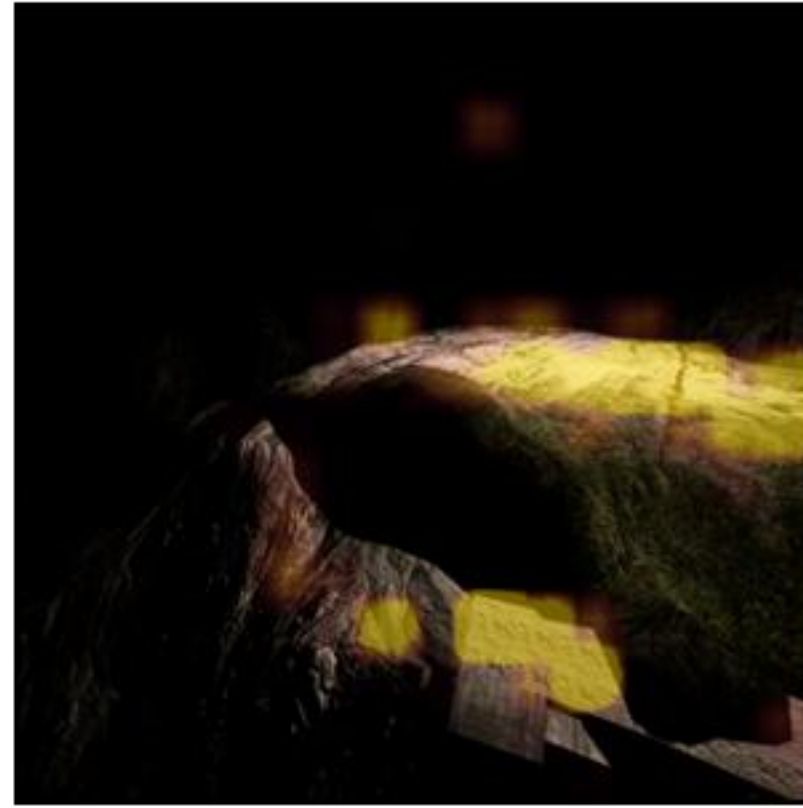
Heatmap Analysis 2: scene Image142

Each panel is the same scene under a different GT light direction. Heatmap target: predicted direction logit.



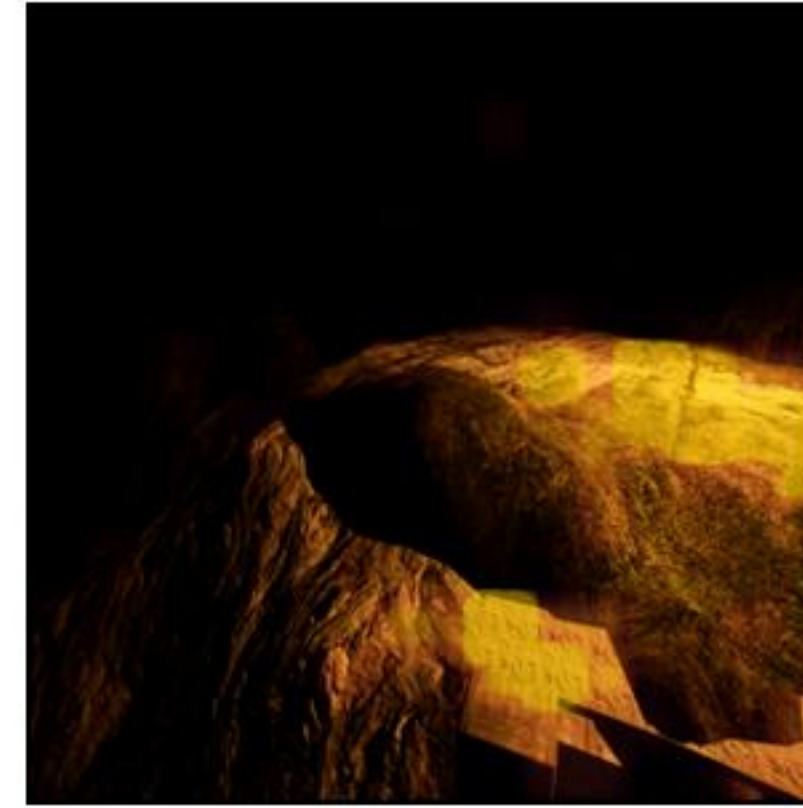
GT: N (0 deg)
Pred: N (0.0 deg)
error: 0.0 deg
cos: 1.0000 conf: 1.000

Image142_N_3500.png



GT: NE (45 deg)
Pred: NE (45.0 deg)
error: 0.0 deg
cos: 1.0000 conf: 1.000

Image142_NE_5500.png



GT: E (90 deg)
Pred: E (90.0 deg)
error: 0.0 deg
cos: 1.0000 conf: 1.000

Image142_E_2500.png



GT: SW (225 deg)
Pred: W (270.0 deg)
error: 45.0 deg
cos: 0.8536 conf: 1.000

Image142_SW_6500.png